

Population genomics of *Tuber aestivum* Vitt. using genome-wide SNPs analysis

Militello Niccolò^{1*}, Lanubile Alessandra^{1,3,4}, Puglisi Edoardo^{2,3,4}, Busconi Matteo^{1,3,4}

¹ Department of Sustainable Crop Production (DIPROVES), Università Cattolica del Sacro Cuore, ² Department for Sustainable Food Process (DISTAS), Università Cattolica del Sacro Cuore ³ National Biodiversity Future Center, Piazza Marina 61, 90133 Palermo (Italy) ⁴ Centre for Research on Biodiversity and Ancient DNA (BioDNA) Via E. Parmense, 84 29122 Piacenza (Italy) *presenting author

Background

- True truffle (*Tuber spp.*) are ascomycete fungi of great ecological and economical importance. They form ectomycorrhizae with different plant species, contributing to ecosystems preservation, and are prized in different national cuisines.
- In recent years interest in summer truffle (*Tuber aestivum* Vitt.) is increasing in Italy, thanks to its subtle aroma, wide distribution and adaptability, especially in the Apennines where it contributes to the local economy and could contribute to woodlands recovery.
- A deep understanding of population genomics in summer truffle is in its early stages. Populations genomics studies are valuable to investigate population structure to understand differences and relatedness across truffle populations, understand population fitness and potentially improve cultivation practices.

Matherial and methods

- 96 samples of summer truffles, collected from ten Italian regions: Apulia, Basilicata, Calabria, Campania, Emilia-Romagna, Lazio, Marche, Piedmont, Tuscany and Umbria were analyzed to obtain genomic profiles. (Figure 1)
- The ascocarps were collected and cleaned to remove any soil contaminants.
- DNA was extracted from the gleba and analysed through a whole genome sequencing (WGS) approach with paired-end short reads.



Figure 1 Geographical Distribution of *Tuber aestivum* collected (created with mapchart.net)

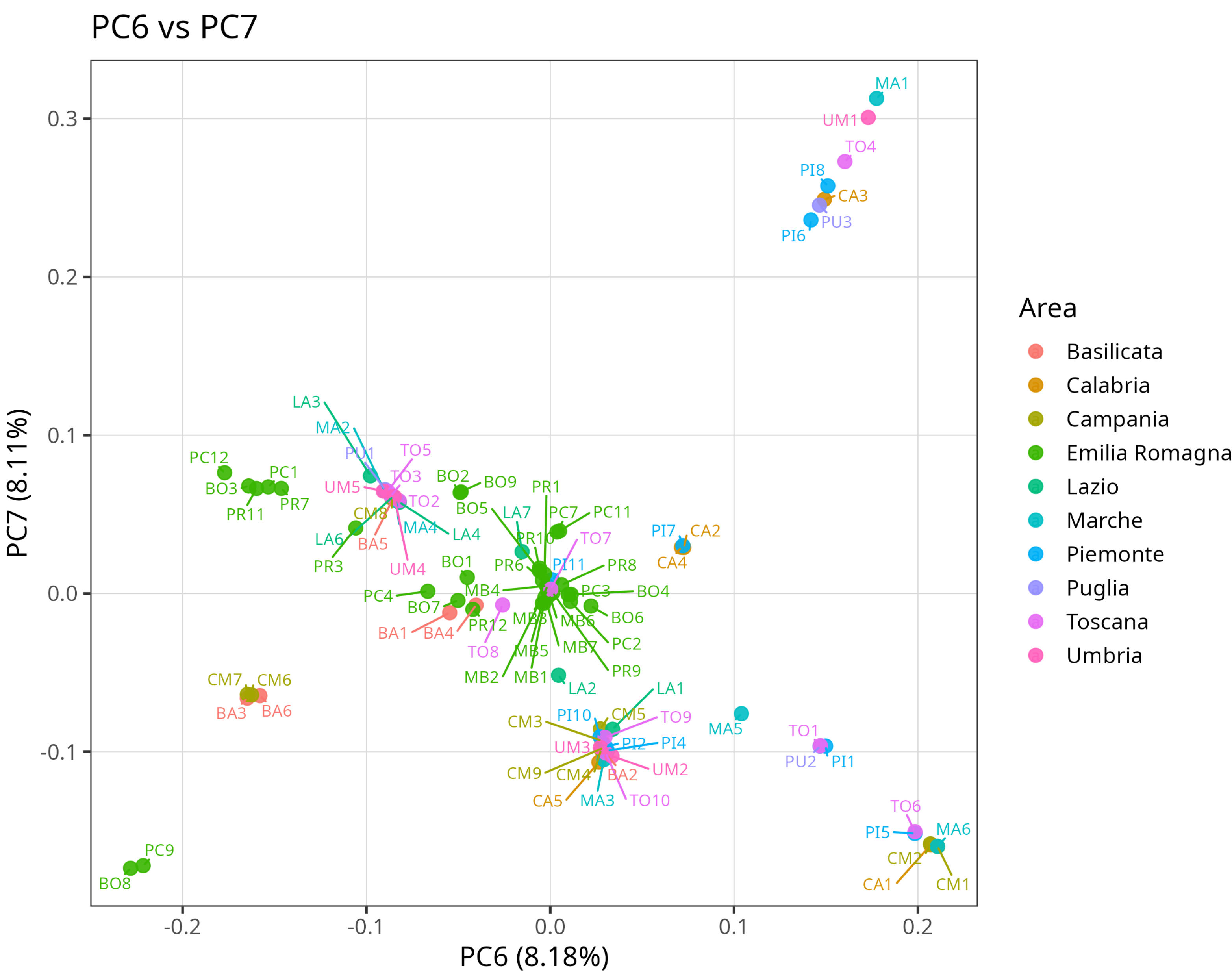


Figure 2 Principal Component Analysis plot of the *Tuber aestivum* population

Results

- PCA (Principal Component Analysis) based on quality-filtered SNPs (Single Nucleotide Polymorphisms) analysis suggested no clear correlation between the haplotype grouping and the geographic origin of the samples, with a core group representing the major genetic variation.
- Some small groups with similar haplotypes and some rare/unique haplotypes could be identified. In addition, a slight geographic structuring along a north-south axis could be hypothesized. (Figure 2)

Summary and prospective

- This study is part of a broader project that aims to use genomics and metagenomic approaches to investigate the genetic structure and diversity of *T. aestivum* and to explore potential correlations between genetic structure, microbiome composition, and environmental parameters to provide data useful to future *Tuber aestivum* biodiversity conservation efforts.
- This poster provides a preliminary genomic characterization of the genetic diversity of Italian *Tuber aestivum* population. Additional analyses are currently ongoing to further explore and characterize genetic variability.

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